HINTS 4 Cycle 3 History Document

April 2021

IMPORTANT! 2021 Data Update

An error occurred where 5-year American Community Survey (ACS) estimates were used as the source of the population totals used in the calibration step of the weighting. The correct population should have been the 1-year ACS estimates. The 5-year estimates are based on an average of the ACS for the previous 5 years, while the 1-year estimates are based on the results of the ACS for that particular year. The HINTS estimates affected most by this error are population totals or counts (e.g., the total number of adults who have searched for information about cancer from any source). These totals will be, on average, about 2 percent lower for the 5-year estimates than if the 1-year estimates were used.

Linear and logistic regressions using the incorrect weights will be affected less than population totals because the error is in both the numerator and the denominator, which will tend to cancel the error out. Several different types of analyses, which compare results using the weights with the error and a corrected set of weights, were completed to test for differences. These involved looking at percentages, regression estimates, and trends. None of these analyses were substantively different when using the corrected weights. Virtually all resulted in percentages, regression coefficients, and significance tests that did not differ at the first decimal place.

The advice to users who have completed analyses using HINTS 4 Cycle 3 but not published yet is to rerun the analyses with the correct weights found in this April 2021 data package downloaded from the HINTS website. For results that have already been published, it is not advised to do anything except in two scenarios:

1. If the results rely on reporting population counts or totals.
2. If a small change in the statistical significance of a result would affect your conclusions. For example, if the result is based on a result that is significant close to a 5% level (if that is the criteria used in the analysis).

In both cases, it is advised to rerun the analysis and decide if the results differ enough to merit reporting an erratum to the journal.

Data Editing

The following variables were identified to contain invalid or unusual values. Those values were replaced with negative value of -4, “Unreadable or Non-conforming numeric response”, negative value of -9, “Missing data (Not Ascertained)” or reasonable regular values.

MailHHAdults: 2. Including yourself, how many people age 18 or older live in this household?

Two respondents had the value of 0, which was replaced by 1.

Height_Feet: F5. (E4.) About how tall are you without shoes? Feet:

Two respondents had a height value that was less than 3 feet. Both values were replaced by -4.

Height_Inches: F5. (E4.) About how tall are you without shoes? Inches:

One hundred and fifty nine respondents had inches of -9, “Missing data (Not Ascertained)”, or -4, “Unreadable or Non-conforming numeric response”, which were replaced by 0. The variable Height_Feet is recommended for determining validity of full height.

Weight: F6. (E5.) About how much do you weigh, in pounds, without shoes?

Two respondents had weights between 0 and 35, which were replaced by -4.

YearCameToUSA: O8. (L8.) In what year did you come to live in the United States?

Fifteen respondents had values between 1 and 99, which were replaced by -4.

SelfAge: O13. (L13.) Starting with yourself, please mark the gender, and write in the age and month of birth for each adult 18 years of age or older living at this address. Self: Age.

Five respondents had age less than 18, which were replaced by -4.

HHAdultAge2: O13. (L13.) Starting with yourself, please mark the gender, and write in the age and month of birth for each adult 18 years of age or older living at this address. Adult 2: Age

Forty three respondents had Adult 2 age between 1 and 17. All their ages were replaced by -4.

HHAdultAge3: O13. (L13.) Starting with yourself, please mark the gender, and write in the age and month of birth for each adult 18 years of age or older living at this address. Adult 3: Age

One hundred and thirty six respondents had Adult 3 age between 0 and 17. All their ages were replaced by -4.

HHAdultAge4: O13. (L13.) Starting with yourself, please mark the gender, and write in the age and month of birth for each adult 18 years of age or older living at this address. Adult 4: Age

One hundred and eighteen respondents had Adult 4 age between 0 and 17. All their ages were replaced by -4.

HHAdultAge5: O13. (L13.) Starting with yourself, please mark the gender, and write in the age and month of birth for each adult 18 years of age or older living at this address. Adult 5: Age

Forty three respondents had Adult 5 age between 0 and 17. All their ages were replaced by -4.

R_HHAdults: Reconciled number of adults in household

Ten respondents had missing values (.), which were replaced by -9.

Totalhousehold: 012. Including yourself, how many people live in your household?

Thirty seven respondents had Totalhousehold of 0, which were replaced by 1.

SharedRes_NotHad: F4. If you had a genetic test, with whom did you personally share the results?

One hundred and three respondents selected a genetic test they had had, but also had SharedRes_NotHad=1. All values of SharedRes_NotHad were replaced by -4 for these respondents.

Standard Recode

Standard recode/derived variables are listed below.

AgeGrpA: -->AgeGrpA. (Age Recode -- 4 Levels)

The Age variable was re-coded into 4 categories: 18-34; 35-39; 40-44; 45+. The original negative values were carried over.

AgeGrpB: -->AgeGrpB. (Age Recode -- 5 Levels)

The Age variable was re-coded into 5 categories: 18-34; 35-49; 50-64; 65-74; 75+. The original negative values were carried over.

EducA: -->EducA. What is the highest level of school you completed? (Education Recode -- 4 Levels)

The Education variable was re-coded into 4 categories: Less than High School; High School Graduate; Some College; College Graduate or More. The original negative values were carried over.

EducB:-->EducB. What is the highest level of school you completed? (Education Recode -- 5 Levels)

The Education variable was re-coded into 5 categories: Less than High School; High School Graduate; Some College; Bachelor's Degree; Post-Baccalaureate Degree. The original negative values were carried over.

RaceEthn:-->Race/Ethnicity. (Hisp_Cat and Race_Cat2 Recode -- 7 Levels)

The RaceEthn was created with Hisp_Cat and Race_Cat2 variables. The RaceEthn has 7 categories: Hispanic; Non-Hispanic White; Non-Hispanic Black or African American; Non-Hispanic American Indian or Alaska Native; Non-Hispanic Asian; Non-Hispanic Native Hawaiian or other Pacific Islander; Non-Hispanic Multiple Races Mentioned. If Hisp_Cat had value of 10, "Not Hispanic", and Race_Cat2 had value of -9, "Missing data (Not Ascertained)", the RaceEthn was assigned with value of -9. The RaceEthn was assigned with value of -9 if Hispanic=-9.

HHInc:-->HHInc. What is your {combined} annual household income? (IncomeRanges Recode -- 5 Levels)

The IncomeRanges variable was re-coded into 5 categories: Less than $20,000; $20,000 to < $35,000; $35,000 to < $50,000; $50,000 to < $75,000; $75,000 or more. The original negative values were carried over.

BMI:-->BMI. Body Mass Index (Weight*703)/(Height in inches**2)

The BMI variable was created with weight in pounds and height in inches. If feet of height or weight had value of -9, “Missing data (Not Ascertained)” but neither had value of -4, “Unreadable or Non-conforming numeric response”, the BMI was assigned to -9. If feet of height or weight had value of -4, “Unreadable or Non-conforming numeric response”, the BMI was assigned to -4.

AgeDX:-->AgeDX. At what age were you diagnosed with cancer? (WhenDiagnosedCancer Recode)

The variable AgeDX is a copy of variable WhenDiagnosedCancer.

TimeSinceDX:-->TimeSinceDX. How long ago were you diagnosed with cancer? (WhenDiagnosedCancer Age Recode)

The variable TimeSinceDX was created with EverHadCancer, WhenDiagnosedCancer and Age variables. The variable TimeSinceDX has 4 categories: Less than 1 Year since DX; 2-5 Years since DX; 6-10 Years since DX; 11+ Years since DX. If the variable EverHadCancer had value of 1 and either Age or WhenDiagnosedCancer had value of -9, “Missing data (Not Ascertained)”, the TimeSinceDX was assigned to -9. If the variable EverHadCancer had value of 1 and WhenDiagnosedCancer is greater than Age, the TimeSinceDX was assigned to -4. If the variable EverHadCancer had value of -9, the TimeSinceDX was assigned to -6. If the variable EverHadCancer had value of 2 and WhenDiagnosedCancer had value of -2, the TimeSinceDX was assigned to -4. If the variable EverHadCancer had value of 2 and WhenDiagnosedCancer had value of -1, the TimeSinceDX was assigned to -1. .

smokeStat:-->smokeStat. Smoking Status (Smoke100 and SmokeNow Recode)

The variable smokeStat was created with Smoke100 and SmokeNow variables. The variable smokeStat has 3 categories: Current; Former; Never. If Smoke100 had value of 1 and SmokeNow had value of -5, “Multiple responses selected in error”, the smokeStat was assigned to -4. If Smoke100 had value of 1 and SmokeNow had value of -9, “Missing data (Not Ascertained)”, the smokeStat was assigned to -9. If Smoke100 had value of -9, “Missing data (Not Ascertained)”, the smokeStat was assigned to -6.

Label Editing

Labels were created for the following recoded variables: AgeGrpA, AgeGrpB, EducA, EducB, RaceEthn, HHInc, BMI, AgeDX, TimeSinceDX, and smokeStat.

Format Editing

Formats Added for Standard Recode Variables

The formats AgeGrpA, AgeGrpB, EducA, EducB, RaceEthn, HHInc, BMI, AgeDX, TimeSinceDX, smokeStat and PHQ4Total were created and assigned to the variables AgeGrpA, AgeGrpB, EducA, EducB, RaceEthn, HHInc, BMI, AgeDX, TimeSinceDX, smokeStat and PHQ4, respectively.

Formats Modified for Certain Variables

All skip patterns in formats were modified (i.e. all instructions to skip questions were deleted). The modified formats are: ACTIVED, ADULTSI, BORNINU, CONSIDE, DRSHOUL, EVERHADC, EVERHAF, FREQGOP, HADTESG, HADTESH, HEARDGE, HEARDHP, INTERNF, LASTMED, OCCUPAF, SEEKHEA, SMOKENO, STRONGF, STRONGG, TIMESMO, USEINTE, WHERESE, and WHERESF.

Imputation of Income Variable

The income variable (IncomeRanges) has relatively higher percentage (14% for un-weighted percentage or 11.6% for weighted percentage) of missing values. This variable was imputed via PROC IMPUTE in SUDAAN. The imputation class variables are: Education (O6), RaceEthn (standard recode), RentOrOwn (O15), BornInUSA (O7) and SpeakEnglish (O9). Since the variable SpeakEnglish was asked for people who were born outside USA (BornInUSA=2), the variable SpeakEnglish was declared after BornInUSA in imputation class statement. The copy variables of the imputation class variables and income variable were created, where the missing values were appropriately coded. The copy variables are used for the imputation. The imputed values were saved in a new variable IncomeRanges_IMP.

SAS Code for Data Editing

```
* Recode MailHHAdults of 0 to 1
*;
      if MailHHAdults=0 then
            MailHHAdults = 1;

* Recode Height_Feet for Anyone who reports height as 3 feet or less
or as 8 or more              *;
      if 0<=Height_Feet<=3 or Height_Feet>=8 then Height_Feet = -4;

* Recode Height_Inches of -9 and -4 to 0
*;
      if Height_Inches <0 then    Height_Inches = 0;

* Recode Weight less than or equal to 35 to Unreadable or Non-
conforming numeric response          *;
```

```
     if 0<=Weight <=35 then  Weight = -4;

* Recode YearCameToUSA between 1 and 99 to Unreadable or Non-
conforming numeric response          *;

     if 1 <= YearCameToUSA <= 99 then   YearCameToUSA = -4;

* Recode SelfAge between 0 and 17 to Unreadable or Non-conforming
numeric response                *;

     if 0 <= SelfAge < 18 then   SelfAge = -4;

* Recode HHAdultAge2-HHAdultAge5 between 0 and 17 to Unreadable or
Non-conforming numeric response*;
     array a(*) HHAdultAge2-HHAdultAge5;

     do I = 1 to dim(a);
          if 0 <= a(I) < 18 then
               a(I) = -4;
     end;

* Recode R_HHAdults of missing (.) to -9
*;
     if missing(R_HHAdults) = 1 then
          R_HHAdults = -9;

* Recode Totalhousehold of 0 to 1;

     if Totalhousehold=0 then Totalhousehold=1;


* Recode SharedRes_NotHad to -4 if HadTest_Cat is 1,2,3,4,5,6,10,or 91
AND SharedRed_NotHad is 1;

     if HadTest_Cat in (1, 2, 3, 4, 5, 6, 10, 91) and SharedRes_NotHad
=1 then SharedRes_NotHad=-4;
```

SAS Code for Standard Recode

```
     if 18 <= Age <= 34 then
          AgeGrpA = 1;
     else if 35 <= Age <= 39 then
          AgeGrpA = 2;
     else if 40 <= Age <= 44 then
          AgeGrpA = 3;
     else if 45 <= Age then
          AgeGrpA = 4;
     else if Age in (-1, -4, -9) then
```

```
            AgeGrpA = Age;
      label AgeGrpA = '-->AgeGrpA. (Age Recode -- 4 Levels)';

      if 18 <= Age <= 34 then
            AgeGrpB = 1;
      else if 35 <= Age <= 49 then
            AgeGrpB = 2;
      else if 50 <= Age <= 64 then
            AgeGrpB = 3;
      else if 65 <= Age <= 74 then
            AgeGrpB = 4;
      else if 75 <= Age then
            AgeGrpB = 5;
      else if Age in (-1, -4, -9) then
            AgeGrpB = Age;
      label AgeGrpB = '-->AgeGrpB. (Age Recode -- 5 Levels)';

      if Education in (1, 2) then
            EducA = 1;
      else if Education in (3) then
            EducA = 2;
      else if Education in (4, 5) then
            EducA = 3;
      else if Education in (6, 7) then
            EducA = 4;
      else if Education in (-1, -4, -9) then
            EducA = Education;
      label EducA = '-->EducA.  What is the highest level of school you
completed? (Education Recode -- 4 Levels)';

      if Education in (1, 2) then
            EducB = 1;
      else if Education in (3) then
            EducB = 2;
      else if Education in (4, 5) then
            EducB = 3;
      else if Education in (6) then
            EducB = 4;
      else if Education in (7) then
            EducB = 5;
      else if Education in (-1, -4, -9) then
            EducB = Education;
      label EducB = '-->EducB.  What is the highest level of school you
completed? (Education Recode -- 5 Levels)';

      if Hisp_Cat in (21, 22, 23, 24, 25) then
            RaceEthn = 1;
      else if Hisp_Cat in (10) then
            do;
                  if Race_Cat2 in (11) then
                        RaceEthn = 2;
                  else if Race_Cat2 in (12) then
```

```
                    RaceEthn = 3;
               else if Race_Cat2 in (14) then
                    RaceEthn = 4;
               else if Race_Cat2 in (31, 32, 33, 34, 35, 36, 37) then
                    RaceEthn = 5;
               else if Race_Cat2 in (51, 52, 53, 54) then
                    RaceEthn = 6;
               else if Race_Cat2 in (16) then
                    RaceEthn = 7;
               else if Race_Cat2 in (-1, -4, -9) then
                    RaceEthn = -9;
          end;
     else if Hisp_Cat in (-1, -4, -9) then
          do;
               RaceEthn = -9;
          end;
     label RaceEthn = '-->Race/Ethnicity. (Hisp_Cat and Race_Cat2
Recode -- 7 Levels)';

     if IncomeRanges in (1, 2, 3) then
          HHInc = 1;
     else if IncomeRanges in (4) then
          HHInc = 2;
     else if IncomeRanges in (5) then
          HHInc = 3;
     else if IncomeRanges in (6) then
          HHInc = 4;
     else if IncomeRanges in (7, 8, 9) then
          HHInc = 5;
     else if IncomeRanges in (-1, -4, -9) then
          HHInc = IncomeRanges;
     label HHInc = '-->HHInc.  What is your {combined} annual
household income? (IncomeRanges Recode -- 5 Levels)';

     if Height_Feet >= 0 and Height_Inches >= 0 and Weight > 0 then
          BMI = (Weight * 703) / ((Height_Feet * 12 +
Height_Inches)**2);
     else if (Height_Feet in (-1, -9) and Weight >=-1 ) or
(Height_Feet >=-1 and Weight in (-1, -9)) or (Height_Feet = -9 and
Weight = -9) then
          BMI = -9;
     else if Height_Feet = -4 or Weight = -4 then
          BMI = -4;
     label BMI = '-->BMI.  Body Mass Index (Weight*703)/(Height in
inches**2)';
     if BMI not in (-4, -9) then
          BMI = round(BMI, 0.1);

     AgeDX = WhenDiagnosedCancer;
     label AgeDX = '-->AgeDX. At what age were you diagnosed with
cancer? (WhenDiagnosedCancer Recode)';
```

```
      if EverHadCancer in (1) then do;
       if Age in (-9) then  TimeSinceDX = Age;
       else if WhenDiagnosedCancer in (-9, -4) and Age >= 18 then
            TimeSinceDX = WhenDiagnosedCancer;
      else if WhenDiagnosedCancer >= 0 and Age >= 18 then do;
         if 0 <= (Age - WhenDiagnosedCancer) <= 1 then
         TimeSinceDX = 1;
         else if 2 <= (Age - WhenDiagnosedCancer) <= 5 then
         TimeSinceDX = 2;
         else if 6 <= (Age - WhenDiagnosedCancer) <= 10 then
         TimeSinceDX = 3;
         else if 11 <= (Age - WhenDiagnosedCancer) then
         TimeSinceDX = 4;
         else if (Age - WhenDiagnosedCancer) < 0 then
         TimeSinceDX = -4;
            end;
      end;
      else if EverHadCancer in (-9) then
            TimeSinceDX = WhenDiagnosedCancer;
      else if EverHadCancer in (2) then
            do;
                  if WhenDiagnosedCancer in (-1) then
                        TimeSinceDX = WhenDiagnosedCancer;
                  else if WhenDiagnosedCancer in (-2) then
                        TimeSinceDX = -4;
            end;
      label TimeSinceDX = '-->TimeSinceDX.  How long ago were you
diagnosed with cancer?  (WhenDiagnosedCancer Age Recode)';

      if Smoke100 in (1) then
            do;
                  if SmokeNow in (1, 2) then
                        smokeStat = 1;
                  else if SmokeNow in (3) then
                        smokeStat = 2;
                  else if SmokeNow in (-5) then
                        smokeStat = -4;
                  else if SmokeNow in (-9) then
                        smokeStat = -9;
            end;
      else if Smoke100 in (2) then
            do;
                  smokeStat = 3;
            end;
      else if Smoke100 in (-9) then
            smokeStat = -6;
      label smokeStat = '-->smokeStat.  Smoking Status (Smoke100 and
SmokeNow Recode)';
```

SAS Code for Label Editing

```
label

            ActiveDutyArmedForces = "O3. (L3.) Have you ever served on
active duty in the U.S. Armed Forces, military Reserves or National
Guard?"

            AverageDailyTVGames = "H6. (F8.) Over the past 30 days, how
many hours per day, on average, did you sit and watch TV or movies,
surf the web, or play computer games?"

            CancerCheckups = "M8. Have you ever received instructions
from a doctor or other health care professional about where you should
return….after completing your cancer treatment?"

            CancerConcernedQuality = "A6c. Based on the results of your
most recent search for information about cancer how much do you agree
or disagree with each of the following statements? Concerned about
Quality."

            CancerFrustrated = "A6b. Based on the results of your most
recent search for information about cancer, how much do you agree or
disagree with each of the following statements? Frustrated."

            CancerLotOfEffort = "A6a. Based on the results of your most
recent search for information about cancer, how much do you agree or
disagree with each of the following statements? Took effort."

            CancerTooHardUnderstand = "A6d. Based on the results of
your most recent search for information about cancer, how much do you
agree or disagree with each of the following statements? Hard to
understand."

            ChanceAskQuestions = "C5a. In the past 12 months, how often
did your health professional: Give you the chance to ask all the
health-related questions you had? "

            ElectCigLessHarm = "J10. Compared to smoking cigarettes,
would you say that electronic cigarettes are…"

            EverHadPSATest = "L8. (I4.) Have you ever had PSA test?"

            ExplainedClearly = "C5e. In the past 12 months, how often
did your health professional: Explain things in a way you could
understand?"
```

FeelingsAddressed = "C5b. In the past 12 months, how often did your health professional: Give the attention you needed to your feelings and emotions?"

Genetics_Diabetes = "E4a. How much do you think genetics determine whether or not a person will develop each of the following conditions? Diabetes or high blood sugar?"

Genetics_HighBP = "E4d. How much do you think genetics determine whether or not a person will develop each of the following conditions? High blood pressure or hypertension?"

HeardGeneticTest = "E2. (A11.) Genetic tests that analyze your DNA, diet and lifestyle are currently being marketed by companies directly to consumers. Have you ever heard or read about these tests?"

HelpUncertainty = "C5g. In the past 12 months, how often did your health professional: Help you deal with feelings of uncertainty aboutyour health or health care?"

InvolvedDecisions = "C5c. In the past 12 months, how often did your health professional: Involve you in decisions about your health care as much as you wanted?"

RegExercise_Enjoyment = "H5d. People start or continue exercising regularly for lots of reasons. How much does getting enjoyment reflect why you would start or continue?"

RegExercise_Guilt = "H5c. People start or continue exercising regularly for lots of reasons. How much does feeling guilty when you skip reflect why you would start or continue?"

TimesModerateExercise = "H1. (F5.) In a typical week, how many days do you do any physical activity of at least moderate intensity?"

TimesStrengthTraining = "H3. (F7.) In a typical week, how many days do you do leisure-time physical activities specifically designed to strengthen your muscles?"

UnderstoodNextSteps = "C5d. In the past 12 months how often did your health professional: Make sure you understood the things you needed to do to take care of your health?"

EverTalkedHPVShot = "L1. Has a doctor or other health care professional ever talked with you about the HPV shot or vaccine?"

```
            HeardOfClinicalTrial = "E1. Clinical trials are research
studies that involve people. Have you ever heard of a clinical trial?"

            SpentEnoughTime = "C5f. In the past 12 months, how often
did your health professional:  Spend enough time with you?"

            ;
```

SAS Code for Format Editing

SAS Code for Formats Added for Standard Recode Variables

```
      value AgeGrpA 1 = '18-34'
                    2 = '35-39'
                    3 = '40-44'
                    4 = '45+'
                    -4 = 'Unreadable or Nonconforming Numeric
Response'
                    -9 = 'Missing Data (Not Ascertained)'
                    ;

      value AgeGrpB 1 = '18-34'
                    2 = '35-49'
                    3 = '50-64'
                    4 = '65-74'
                    5 = '75+'
                    -4 = 'Unreadable or Nonconforming Numeric
Response'
                    -9 = 'Missing Data (Not Ascertained)'
                    ;

      value EducA 1 = 'Less than High School'
                    2 = 'High School Graduate'
                    3 = 'Some College'
                    4 = 'College Graduate or More'
                    -9 = 'Missing Data (Not Ascertained)'
                    ;

      value EducB 1 = 'Less than High School'
                    2 = 'High School Graduate'
                    3 = 'Some College'
                    4 = "Bachelor's Degree"
                    5 = 'Post-Baccalaureate Degree'
                    -9 = 'Missing Data (Not Ascertained)'
                    ;

      value RaceEthn 1 = 'Hispanic'
                       2 = 'Non-Hispanic White'
```

```
                          3 = 'Non-Hispanic Black or African
American'
                          4 = 'Non-Hispanic American Indian or Alaska
Native'
                          5 = 'Non-Hispanic Asian'
                          6 = 'Non-Hispanic Native Hawaiian or other
Pacific Islander'
                          7 = 'Non-Hispanic Multiple Races Mentioned'
                          -4 = 'Unreadable or Nonconforming Numeric
Response'
                          -9 = 'Missing Data (Not Ascertained)'
                          ;

      value HHInc 1 = 'Less than $20,000'
                  2 = '$20,000 to < $35,000'
                  3 = '$35,000 to < $50,000'
                  4 = '$50,000 to < $75,000'
                  5 = '$75,000 or More'
                  -5 = 'Multiple Responses Selected in Error'
                  -9 = 'Missing Data (Not Ascertained)'
                  ;

      value BMI -4 = 'Unreadable or Nonconforming Numeric Response'
                -9 = 'Missing Data (Not Ascertained)'
                ;

      value AgeDX -1 = 'Inapplicable, coded 2 in EverHadCancer'
                  -2 = 'Question Answered in Error (Commission
Error)'
                  -6 = 'Missing Data (Filter Missing)'
                  -9 = 'Missing Data (Not Ascertained)'
                  ;

      value TimeSinceDX 1 = 'Less than 1 Yr Since DX'
                        2 = '2-5 Yrs Since DX'
                        3 = '6-10 Yrs Since DX'
                        4 = '11+ Yrs Since DX'
                        -1 = 'Inapplicable, coded 2 in
EverHadCancer'
                        -4 = 'Unreadable or Nonconforming Numeric
Response'
                        -6 = 'Missing Data (Filter Missing), coded
-9 in EverHadCancer'
                        -9 = 'Missing Data (Not Ascertained)'
                        ;

      value smokeStat 1 = 'Current'
                        2 = 'Former'
                        3 = 'Never'
                        -4 = 'Unreadable or Nonconforming Numeric
Response'
```

```
                              -6 = 'Missing Data (Filter Missing), coded
-9 in Smoke100'
                              -9 = 'Missing Data (Not Ascertained)'
                              ;


      format      AgeGrpA AgeGrpA.
                  AgeGrpB AgeGrpB.
                  EducA EducA.
                  EducB EducB.
                  RaceEthn RaceEthn.
                  HHInc HHInc.
                  BMI BMI.
                  AgeDX AgeDX.
                  TimeSinceDX TimeSinceDX.
                  smokeStat smokeStat.
                  ;
```

SAS Code for Imputation of Income Variable

```
* Impute IncomeRanges via PROC HOTDECK
*;
data HINTS4CYCLE3;
      set HINTS4CYCLE3;

      COPY_Education = Education;
      if COPY_Education in (-9) then
            COPY_Education = .;

      COPY_RaceEthn = RaceEthn;
      if COPY_RaceEthn in (-9) then
            COPY_RaceEthn = .;

      COPY_RentOrOwn = RentOrOwn;
      if COPY_RentOrOwn in (-5, -9) then
            COPY_RentOrOwn = .;

      COPY_SpeakEnglish = SpeakEnglish;
      if COPY_SpeakEnglish in (-1, -2, -5, -6, -9) then
            COPY_SpeakEnglish = .;

      COPY_BornInUSA = BornInUSA;
      if COPY_BornInUSA in (-9) then
            COPY_BornInUSA = .;

      COPY_IncomeRanges = IncomeRanges;
      if COPY_IncomeRanges in (-9) then
```

```
            COPY_IncomeRanges = .;

      ID = _N_;

      format COPY_Education Educati. COPY_RaceEthn RaceEthn.
COPY_RentOrOwn RentOrO.
                  COPY_SpeakEnglish SpeakEn. COPY_BornInUSA BornInU.;
run;

proc freq data=HINTS4CYCLE3;
      tables COPY_Education*Education / list missing;
      tables COPY_RaceEthn*RaceEthn / list missing;
      tables COPY_RentOrOwn*RentOrOwn / list missing;
      tables COPY_SpeakEnglish*SpeakEnglish / list missing;
      tables COPY_BornInUSA*BornInUSA / list missing;
      tables COPY_IncomeRanges*IncomeRanges / list missing;
run;

proc impute data=HINTS4CYCLE3 method=wshd notsorted;
      weight person_finwt0;
      impvar COPY_IncomeRanges;
      impby COPY_Education COPY_RaceEthn COPY_RentOrOwn COPY_BornInUSA
COPY_SpeakEnglish;
      impname COPY_IncomeRanges="IncomeRanges_IMP";
      impid ID;
      output IMPID IMPBY IMPUTEVAL / filename=impute1 replace;
run;

proc freq data=impute1;
      tables IncomeRanges_IMP / missing;
run;

proc contents data=impute1;
run;

proc sort data=HINTS4CYCLE3;
      by ID;
run;

proc sort data=impute1 (keep=ID IncomeRanges_IMP);
      by ID;
run;

data HINTS4CYCLE3;
      merge HINTS4CYCLE3 (in=A) impute1 (in=B);
      by ID;

      if A = 1 and B = 1;
run;

data _null_;
      set HINTS4CYCLE3;
```

```
      if IncomeRanges not in (-9) and COPY_IncomeRanges ^=
IncomeRanges_IMP then
            put ID IncomeRanges COPY_IncomeRanges IncomeRanges_IMP;
run;

data HINTS4CYCLE3;
      set HINTS4CYCLE3;

      if missing(IncomeRanges_IMP) = 1 then
            IncomeRanges_IMP = IncomeRanges;
      label IncomeRanges_IMP = '-->IncomeRanges_IMP.  Imputed
IncomeRanges variable via PROC HOTDECK in SUDAAN';
      format IncomeRanges_IMP IncomeR.;

      drop COPY_Education COPY_RaceEthn COPY_RentOrOwn
COPY_SpeakEnglish COPY_BornInUSA
            ID
            COPY_IncomeRanges;

      format MultiOcc;
run;
```